

TripJack Assignment

Loan Data Analysis

-- 1: Average Loan Amount by Credit History Status

```
SELECT Credit_History, round(AVG(LoanAmount),2) AS Average_Loan_Amount
FROM loan_applications
GROUP BY Credit_History;
```

```
23      -- Analyses
24
25      -- 1: Average Loan Amount by Credit History Status
26
27      SELECT Credit_History, round(AVG(LoanAmount),2) AS Average_Loan_Amount
28      FROM loan_applications
29      GROUP BY Credit_History;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
Credit_History	Average_Loan_Amount			
1	104.47			
0	108.73			

2: Count of Approved vs. Rejected Loans

```
SELECT Loan_Status, COUNT(*) AS Total_Count
FROM loan_applications
GROUP BY Loan_Status;
```

```

31
32 -- 2: Count of Approved vs. Rejected Loans
33
34 • SELECT Loan_Status, COUNT(*) AS Total_Count
35 FROM loan_applications
36 GROUP BY Loan_Status;
37
38
39
40
41
42

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Loan_Status	Total_Count
▶	N	101
	Y	239

-- 3. Distribution of Loans by Property Area:

```

SELECT Property_Area, COUNT(*) AS Total_Loans
FROM loan_applications
GROUP BY Property_Area;

```

```

37
38 -- 3.Distribution of Loans by Property Area:
39
40 • SELECT Property_Area, COUNT(*) AS Total_Loans
41 FROM loan_applications
42 GROUP BY Property_Area;
43
44

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Property_Area	Total_Loans
▶	Rural	95
	Urban	108
	Semiurban	137

-- 4. Average Applicant Income by Education Level:

```
SELECT Education, AVG(ApplicantIncome) AS Average_Income
FROM loan_applications
GROUP BY Education;
```

```
44
45 -- 4.Average Applicant Income by Education Level:
46
47 • SELECT Education, AVG(ApplicantIncome) AS Average_Income
48 FROM loan_applications
49 GROUP BY Education;
```

51  Loan Approval Rate by Marital Status:

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
Education	Average_Income			
Graduate	3630.09126984127			
Not Graduate	3494.784090909091			

-- 5.Loan Approval Rate by Marital Status:

```
SELECT Married,
SUM(CASE WHEN Loan_Status = 'Y' THEN 1 ELSE 0 END) AS Approved_Loans,
SUM(CASE WHEN Loan_Status = 'N' THEN 1 ELSE 0 END) AS Rejected_Loans
FROM loan_applications
GROUP BY Married;
```

```
50
51 -- 5.Loan Approval Rate by Marital Status:
52
53 • SELECT Married,
54 SUM(CASE WHEN Loan_Status = 'Y' THEN 1 ELSE 0 END) AS Approved_Loans,
55 SUM(CASE WHEN Loan_Status = 'N' THEN 1 ELSE 0 END) AS Rejected_Loans
56 FROM loan_applications
57 GROUP BY Married;
```

59  Comparison of Average Loan Amount for Self-Employed vs. Non-Self-Employed Applicant:

Married	Approved_Loans	Rejected_Loans
Yes	149	52
No	90	49

-- 6. Comparison of Average Loan Amount for Self-Employed vs. Non-Self-Employed Applicants:

```
SELECT Self_Employed, AVG(LoanAmount) AS Average_Loan_Amount
FROM loan_applications
GROUP BY Self_Employed;
```

The screenshot shows a SQL query editor with the following code:

```
58
59 -- 6. Comparison of Average Loan Amount for Self-Employed vs. Non-Self-Employed Applicants:
60
61 • SELECT Self_Employed, round(AVG(LoanAmount),2) AS Average_Loan_Amount
62 FROM loan_applications
63 GROUP BY Self_Employed;
```

Below the query editor is a 'Result Grid' showing the results of the query:

Self_Employed	Average_Loan_Amount
No	104.73
Yes	106.93
	108.1

-- 7. Loan Approval Rate by Dependents

```
SELECT Dependents,
SUM(CASE WHEN Loan_Status = 'Y' THEN 1 ELSE 0 END) AS Approved_Loans,
SUM(CASE WHEN Loan_Status = 'N' THEN 1 ELSE 0 END) AS Rejected_Loans
FROM loan_applications
GROUP BY Dependents;
```

The screenshot shows a SQL query editor with the following code:

```
66 -- 7. Loan Approval Rate by Dependents
67 • SELECT Dependents,
68 SUM(CASE WHEN Loan_Status = 'Y' THEN 1 ELSE 0 END) AS Approved_Loans,
69 SUM(CASE WHEN Loan_Status = 'N' THEN 1 ELSE 0 END) AS Rejected_Loans
70 FROM loan_applications
71 GROUP BY Dependents;
```

Below the query editor is a 'Result Grid' showing the results of the query:

Dependents	Approved_Loans	Rejected_Loans
1	32	14
0	146	62
2	39	15
3+	19	6
	3	4

-- 8. Loan Approval Rate by Gender

```
SELECT Gender,  
  
SUM(CASE WHEN Loan_Status = 'Y' THEN 1 ELSE 0 END) AS Approved_Loans,  
  
SUM(CASE WHEN Loan_Status = 'N' THEN 1 ELSE 0 END) AS Rejected_Loans  
  
FROM loan_applications  
  
GROUP BY Gender;
```

The screenshot shows a SQL query editor with the following code:

```
73  
74 -- 8. Loan Approval Rate by Gender  
75 • SELECT Gender,  
76     SUM(CASE WHEN Loan_Status = 'Y' THEN 1 ELSE 0 END) AS Approved_Loans,  
77     SUM(CASE WHEN Loan_Status = 'N' THEN 1 ELSE 0 END) AS Rejected_Loans  
78 FROM loan_applications  
79 GROUP BY Gender;  
80  
81
```

Below the editor is a 'Result Grid' with the following data:

	Gender	Approved_Loans	Rejected_Loans
▶	Male	191	72
	Female	47	25
		1	4

-- 9. Average Coapplicant Income by Loan Status

```
SELECT Loan_Status, round(AVG(CoapplicantIncome),2) AS Average_Coapplicant_Income  
  
FROM loan_applications  
  
GROUP BY Loan_Status;
```

The screenshot shows a SQL query editor with the following code:

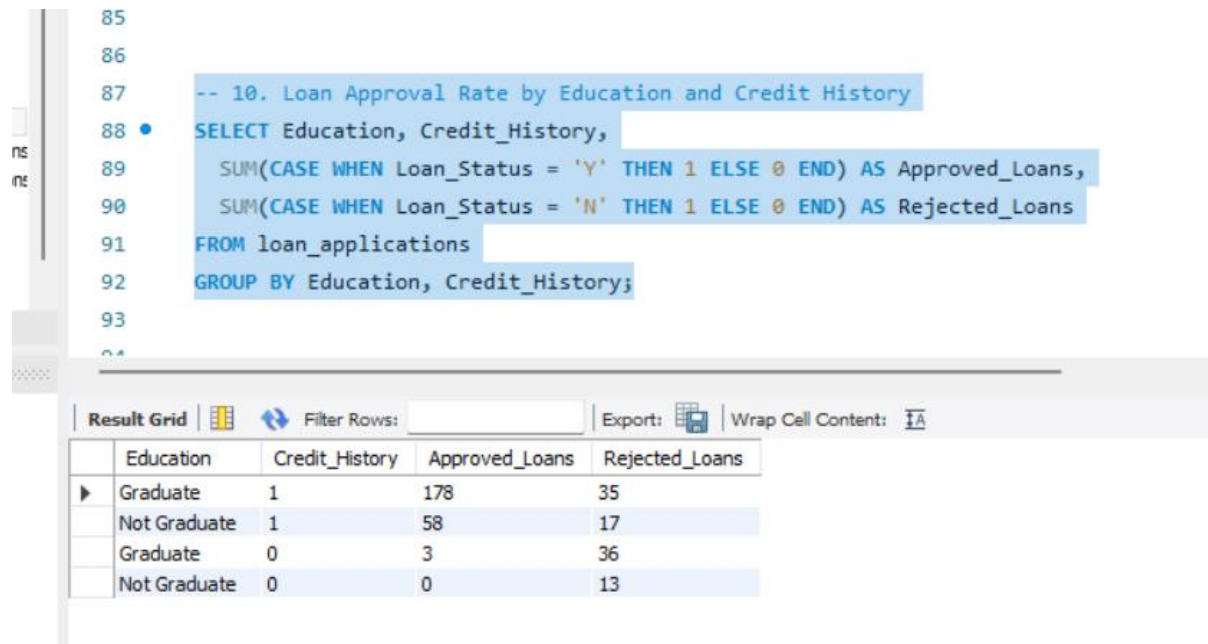
```
80  
81 -- 9. Average Coapplicant Income by Loan Status  
82 • SELECT Loan_Status, round(AVG(CoapplicantIncome),2) AS Average_Coapplicant_Income  
83 FROM loan_applications  
84 GROUP BY Loan_Status;  
85  
86
```

Below the editor is a 'Result Grid' with the following data:

	Loan_Status	Average_Coapplicant_Income
▶	N	1222.76
	Y	1271.09

-- 10. Loan Approval Rate by Education and Credit History

```
SELECT Education, Credit_History,  
       SUM(CASE WHEN Loan_Status = 'Y' THEN 1 ELSE 0 END) AS Approved_Loans,  
       SUM(CASE WHEN Loan_Status = 'N' THEN 1 ELSE 0 END) AS Rejected_Loans  
FROM loan_applications  
GROUP BY Education, Credit_History;
```



The screenshot shows a SQL IDE interface. The query editor on the left contains the SQL code for query 10. The results pane on the right displays a table with the following data:

Education	Credit_History	Approved_Loans	Rejected_Loans
Graduate	1	178	35
Not Graduate	1	58	17
Graduate	0	3	36
Not Graduate	0	0	13

-- 11. Loan Approval Rate by Applicant Income and Credit History

```
SELECT  
CASE  
  WHEN ApplicantIncome <= 3000 THEN '0-3000'  
  WHEN ApplicantIncome > 3000 AND ApplicantIncome <= 6000 THEN '3001-6000'  
  WHEN ApplicantIncome > 6000 AND ApplicantIncome <= 9000 THEN '6001-9000'  
  ELSE '9001 and above'  
END AS Applicant_Income_Range,  
Credit_History,  
COUNT(*) AS Total_Loans,  
SUM(CASE WHEN Loan_Status = 'Y' THEN 1 ELSE 0 END) AS Approved_Loans,  
SUM(CASE WHEN Loan_Status = 'N' THEN 1 ELSE 0 END) AS Rejected_Loans
```

FROM loan_applications

GROUP BY Applicant_Income_Range, Credit_History;

The screenshot shows a SQL IDE interface. On the left, there's a sidebar with a tree view containing 'ects', '56', 'esdb', 'rtis', 'n_analytics', 'a', 'rstore_db', 'ack', 'ables', 'loan_applications', 'loan_applications', 'iews', 'tored Procedures', 'unctions', 'i', 'ion', and 'Schemas'. The main area displays a SQL query starting with a comment '-- 11. Loan Approval Rate by Applicant Income and Credit History'. The query uses a CASE statement to categorize ApplicantIncome into ranges: '0-3000', '3001-6000', '6001-9000', and '9001 and above'. It then calculates Total_Loans, Approved_Loans, and Rejected_Loans for each combination of Applicant_Income_Range and Credit_History. The results are shown in a table with columns: Applicant_Income_Range, Credit_History, Total_Loans, Approved_Loans, and Rejected_Loans. The table has 10 rows of data.

```
95 -- 11. Loan Approval Rate by Applicant Income and Credit History
96 SELECT
97 CASE
98     WHEN ApplicantIncome <= 3000 THEN '0-3000'
99     WHEN ApplicantIncome > 3000 AND ApplicantIncome <= 6000 THEN '3001-6000'
100    WHEN ApplicantIncome > 6000 AND ApplicantIncome <= 9000 THEN '6001-9000'
101    ELSE '9001 and above'
102 END AS Applicant_Income_Range,
103 Credit_History,
104 COUNT(*) AS Total_Loans,
105 SUM(CASE WHEN Loan_Status = 'Y' THEN 1 ELSE 0 END) AS Approved_Loans,
106 SUM(CASE WHEN Loan_Status = 'N' THEN 1 ELSE 0 END) AS Rejected_Loans
107 FROM loan_applications
108 GROUP BY Applicant_Income_Range, Credit_History;
```

Applicant_Income_Range	Credit_History	Total_Loans	Approved_Loans	Rejected_Loans
3001-6000	1	160	128	32
0-3000	1	109	93	16
3001-6000	0	27	1	26
6001-9000	0	4	0	4
0-3000	0	21	2	19
6001-9000	1	17	13	4
9001 and above	1	2	2	0

-- 12. Approval rate based on credit history, applicant income and co-applicant income

SELECT

Credit_History,

ROUND(AVG(CASE WHEN Loan_Status = 'Y' THEN 1.0 ELSE 0.0 END), 2) AS Approval_Rate,

MIN(ApplicantIncome) AS Min_Applicant_Income,

MAX(ApplicantIncome) AS Max_Applicant_Income,

AVG(ApplicantIncome) AS Avg_Applicant_Income,

MIN(CoapplicantIncome) AS Min_Coapplicant_Income,

MAX(CoapplicantIncome) AS Max_Coapplicant_Income,

round(AVG(CoapplicantIncome),2) AS Avg_Coapplicant_Income

FROM loan_applications

GROUP BY Credit_History;

